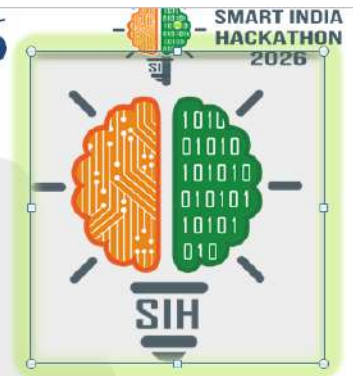


SMART INDIA HACKATHON 2026

- **Problem Statement ID** – SIH26037
- **Problem Statement Title** - Adaptive Path Planning For Autonomous Vehicles On Unstructured Indian Roads
- **Theme** –
- **PS Category** – Software
- **Team ID** –
- **Team Name** – The Horizon



The Horizon

ROAD-SENSE ADAPTIVE PLANNER

lane-independent autonomy stack converts messy road reality
into safe continuously updated driving corridor



4 STEPS

01 PERCEIVE

Fuse camera + LiDAR + radar to local world model

02 PREDICT

Track every road user forecast short-term motion

03 DECIDE

Estimate risk & traversability • choose safest intent

04 REPLAN

Generate collision-free path whenever scene changes

HOW IT ADDRESSES PROBLEM

- **No lane dependency** — uses traversability for open-road no-lane environments
- **Mixed traffic** — handles pedestrians cyclists bikes autos cars animal
- **Sudden events** — risk-aware replanning for cut-ins obstacles unexpected hazards
- **Uncertain scenes** — confidence-gated planning • safe fallback to stop

✧ WHAT MAKES IT NOVEL

**PATH SCORE = SAFETY + TRAVERSABILITY
+ COMFORT + PROGRESS**

Dynamic weights shift with density • speed • risk •
• environment conditions



Density
Adaptive

Real-time
Reweighting

Comfort-Aware

Dynamic Weight Adjustment

TECHNICAL APPROACH

CLOSED-LOOP PERCEPTION • PREDICTION • PLANNING



SENSORS

- Camera • LiDAR
- Radar



FUSION

- Object tracks
- Road edges
- Free-space



PREDICTION

- Kinematics
- Intent cues
- Uncertainty



PLANNER

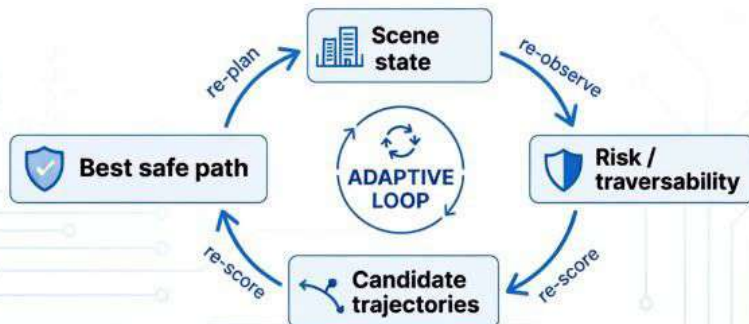
- Risk field
- Candidate paths
- Replanning



CONTROL

- Bicycle model
- Steering speed
- Feedback

ADAPTIVE LOCAL PLANNING LOOP



IMPLEMENTATION STACK



MATLAB Simulink

Core simulation



RoadRunner

Indian road scenarios



Automated Driving Toolbox

Sensor modeling & fusion



Stateflow

Decision logic



Deep Learning Toolbox

Detection & prediction



Navigation Toolbox

Planning utilities

The Horizon

FEASIBILITY & VIABILITY



SMART INDIA
HACKATHON
2026

✓ BUILDABLE NOW • TESTABLE IN SIMULATION



SIMULATION-FIRST

Build complete loop in MATLAB
Simulink before deployment



MODULAR AI

Swap detectors • predictors • planners
independently • failures localized



SCENARIO-DRIVEN

Validate against five required
Indian road situations



VALIDATION MATRIX

- 01 Unmarked village road**
Undefined edges • potholes
- 02 Unsignalized intersection**
Crossing • informal merging
- 03 Highway merge**
Slow • fast • mixed traffic
- 04 Dense market corridor**
Pedestrians • pushcarts
- 05 Sudden cattle crossing**
Mid-path replanning • stop



CHALLENGE MITIGATION

- ✓ **Occlusion**
Multi-sensor redundancy
- ✓ **Erratic motion**
Short-horizon prediction + risk buffer
- ✓ **Noisy perception**
Confidence-aware planning
- ✓ **Computation**
Local horizon + event-triggered replanning

IMPACT AND BENEFITS

FROM LANE-FOLLOWING TO SITUATION-AWARE DRIVING

TRADITIONAL ASSUMPTION



**LANES
OBJECTS
REACT**



SAFETY

Earlier reaction to crossing, merging, unexpected obstacles



LOCAL RELEVANCE

Designed around Indian mixed traffic and road behavior

OUR ADAPTIVE MODEL



**FREE SPACE
RISK
ADAPT**



ROBUSTNESS

Works when lane markings and formal road geometry disappear



SCALABILITY

Same architecture can transfer from simulation to real-world stacks



45 ms
REAL-TIME LATENCY

Fast processing for immediate decision-making



0.9 m/s³
SMOOTHNESS JERK

Low jerk for passenger comfort & stability

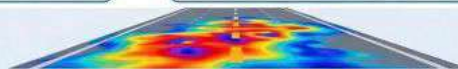


100%
SAFETY GUARANTEE

Reliable safety assurance across scenarios



Lane-following • fixed lanes



Risk heatmap • free-space & dynamic risk

"Don't ask the road to behave like a lane map — make the planner adapt to the road as it is."

KEY REFERENCES & LITERATURE

IEEE Transactions on Intelligent Transportation Systems
Unstructured Environment Navigation in Indian Context

Risk Field Formulations & Dynamic Potential Mapping
for Autonomous Driving in Dense Heterogeneous Traffic

Indian Traffic Edge-Case Dataset Benchmarks &
Digital Twin Simulation Frameworks

SIMULATION & BENCHMARKS

MATLAB

RoadRunner

MathWorks

SIH 2026

